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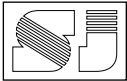


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## Analyses of Aggressiveness, Impulsiveness, and Demographics of the Drivers in Sulaimaniyah City Using Questionnaire Forms

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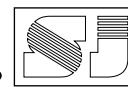
### Abstract

In order to increase traffic safety on our roads, certain forms of behavior and personality traits of passenger car drivers were studied. As an attempt to understand the potential contribution of drivers' impulsiveness and aggressiveness in traffic accidents in Sulaimaniyah City, this study was conducted. The correlation between drivers' impulsiveness and aggressiveness were also explored. Participants, who filled Barratt Impulsiveness Scale (BIS-11) and Aggressive Driving Behavior Questionnaire (ADBQ), were 244 drivers. The male drivers who answered the questionnaires were 143, while female drivers were 101. The results of the statistical analyses showed that male drivers are driving more aggressively than female drivers; as a result, male drivers faced higher numbers of traffic accidents than female drivers. There were no significant differences between male and female drivers regarding drivers' impulsivity. Speeding as a measuring scale of the aggressive driving is significantly correlated with second-order impulsiveness subscales. The attentional and motor impulsiveness subscales are more correlated with the total score of the driving aggressiveness than non-planning subscale. There was moderate correlation between the impulsiveness total score and the overall aggressiveness score. The impulsiveness of the drivers was negatively correlated with the drivers' ages and positively correlated with number of crashes; while the driving aggressiveness was significantly correlated with number of crashes and negatively with gender and age of the drivers. The traffic police in Sulaimaniyah City can benefit from the results of this paper during permitting driving license and enforcement processes.

### 1. Introduction

Road traffic accidents, which cause deaths and injuries of road users, remain a serious problem globally. According to the World Health Organization (WHO) report in 2018, the number of fatal accidents of road users due to road traffic crashes continue to climb, reaching 1.35 million in 2016 <sup>[1]</sup>. Human behavior traits such as aggressiveness, impulsiveness, and other personal

similarity traits such as sensation seeking are the most important behavioral factors in traffic crashes <sup>[2]</sup>. A study shows that a human factor is mainly responsible for around 90 to 95% of traffic crashes <sup>[3]</sup>. Among the human behaviors, one of the major contributors to traffic crashes is aggressive driving. Some other data show that less percentages of crashes that happened due to aggressive driving; for instance, according to 2014 traffic accident data of PennDOT, 55% of all



crashes happened in five counties were caused by aggressive driving. Among all fatalities from traffic crashes, 52% of them were caused by aggressive driving behavior <sup>[4]</sup>.

During the 1990s, the word aggressive driving emerged as a label for a type of drivers whose behavior are dangerous during driving <sup>[5]</sup>. The National Highway Traffic Safety Administration (NHTSA) defines aggressive driving as the operation of a motor vehicle in a manner that endangers or is likely to endanger persons or properties <sup>[5]</sup>. Tasca (2000) illustrated that "a driving behavior is aggressive if it is deliberate, likely to increase the risk of collision, and is motivated by impatience, annoyance, hostility, and/or an attempt to save time <sup>[6]</sup>". Traffic safety communities have attempted to measure aggressive driving by focusing on specific behaviors that the drivers increase the potential of crashes such as flashing lights, honking, verbal threats to other road users, gestures, following too closely, driving at excessive speeds, weaving through traffic, blocking and cutting the road to other vehicles, and running stop lights and signs <sup>[7] [8]</sup>.

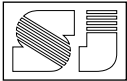
For assessing different aspects of aggressive driving, several scale measures of questionnaires were designed such as the Driver Behavior Questionnaire (DBQ) <sup>[9]</sup>, the Driver Anger Questionnaire (DAEI) <sup>[10]</sup>, the Driving Angry Thoughts Questionnaire (DATQ) <sup>[11]</sup>, the Driver Anger Scale (DAS) and Aggressive Driver Behavior Questionnaire (ADBQ) <sup>[12]</sup>.

Impulsiveness is defined as "swift action without forethought or conscious judgment, behavior without adequate thought, and the tendency to act with less forethought than do most individuals of equal ability and knowledge" <sup>[13]</sup>. Impulsivity, which implies a predisposition toward unplanned and rapid reactions to internal and external stimuli, may leads to unsafe diving behaviors such as not fastening the seat belts, driving under the influence of alcohol or drugs, ignoring traffic control devices, and risky driving <sup>[14]</sup>.

## 2. Literature Review

Several methods of questionnaires have been determined to measure impulsiveness such as Barratt Impulsiveness Scale-11 <sup>[15]</sup>, Urgency, Premeditation, Perseverance, and Sensation Seeking (UPPS) Impulsiveness Behavior Scale <sup>[16]</sup>, and Eysenck Impulsivity Scale-17 <sup>[17]</sup>.

Several studies have been conducted about the effects of human behavior traits on aggressive driving behavior; as well as their effects on traffic safety. Tasca (2000) identified some important factors that affect driving behavior such as driver's age, gender, traffic situation, social situation emotions or aggressiveness, angry mood, and driving skills <sup>[6]</sup>. James and Nahl (2000) described the impacts of emotional condition and necessity of self-regulation on driver behavior. The emotional disorders and insufficient qualification of the driver were the factors for the drivers to be aggressive <sup>[18]</sup>. Lajunen (2001) evaluated the effects of personal variables such as extroversion, neuroticism and psychoticism on traffic crashes. The fatal number due to road crashes was correlated to the extroversion positively, whereas the fatality was correlated to the neuroticism negatively <sup>[19]</sup>. Krahé and Fenske (2002) obtained that there are significant relationships between aggressive driving, Macho personality (hypermasculinity), age, and power of car <sup>[20]</sup>. After assessing influence of the personality dimension "locus of control", which relates to an individual's assumptions concerning responsibility for positive and negative events, on driver behavior by Rudin-Brown and Noy (2002), it was concluded that locus control is an important factor that influences on the drivers' behavior <sup>[21]</sup>; however, the results obtained from the studies of Iversen & Rundmo (2002) and Özkan & Lajunen, (2010) were contradictory <sup>[7] [22]</sup>. Philippe et al. (2009) showed that an obsessive passion for driving was associated with aggressive driving behavior, while harmonious passion was not. Also, the emotion of anger mediated the obsessive-passion/aggressive driving- behavior relationship <sup>[23]</sup>. Abojaradeh et



al. (2014) concluded that the main causes of traffic accidents, injuries, and fatalities are related to driver behavior in Jordan <sup>[24]</sup>. Čabarkapa et al. (2018) compared the levels of impulsiveness and aggressiveness among three considered groups, which were bus, truck, and non-professional drivers. There were significant relationships between drivers' impulsiveness and aggressiveness. Bus and truck drivers had higher levels of impulsiveness compared to non-professional drivers; while the non-professional drivers had the highest level of aggressiveness, followed by truck and bus drivers <sup>[14]</sup>.

Other study assessed the effects of driver behavior on traffic congestion; for instance, the study of Muti et al. (2018) in which the aggressive driving behaviors on Bandung City were categorized into five factors namely improper speed, inattentiveness, display of hostility, impatience, and disobedience of traffic sign/signals. The study showed that different composition of driving behaviors leads to different degree of congestion. In addition, impatience behavior was obtained to be the factor that must be eliminate to decrease the congestion on Bandung City <sup>[25]</sup>.

A review study of Bicakisia and Özakan investigated the relationship between impulsivity and driver behaviors, offences and road traffic accidents through the lenses of characterological perspective. A comprehensive review of the published studies from 1970 to 2014 that addressed the relationship between drivers' impulsiveness and at least one driving related outcome (e.g., a self-report measure of driver behavior) were presented. The studies were separated into four sections based on the driving related outcomes as; (i) aberrant driver behaviors and driving anger/aggression, (ii) driving under the influence, (iii) traffic offences and accidents, (iv) other. Most of the studies reported significant relationships between impulsivity and the driving outcomes <sup>[26]</sup>.

To improve traffic safety by reducing the number of crashes and to increase traffic efficiency, driver behavior traits should be further investigated. This study was conducted as an effort to

understand certain forms of driver behavior and drivers' personality traits in Sulaimaniyah City. The study also evaluates the potential contribution of drivers' impulsiveness and aggressiveness to traffic accidents.

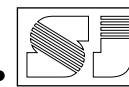
### 3. Methodology

The aggressiveness, impulsiveness, and demographic data of the drivers in Sulaimaniyah City were analyzed by using Barratt Impulsiveness (BIS-11) and Aggressive Driving Behavior Questionnaire (ADBQ) scales. The Barratt impulsiveness scale (BIS-11) consists of 30 questions that further subscale into first and second order factors as shown in Table 1. The questions were scored on a four-point Likert scale (1 = rarely/never, 2 = occasionally, 3 = often, and 4 = almost/always).

The ADBQ, which is shown in Table 2, consists of 20 questions that were filled by drivers.

The ADBQ is further divided into four groups named as anger/aggression, speeding, overt expression, and judgment of others as shown in Table 3. The ADBQ questions were scored at six-point Likert scale (1 = never, 2 = hardly at all, 3 = occasionally, 4 = often, 5 = quite frequently, 6 = nearly all the time).

Among the demographic variables that were used gender and age as a demographic indicator, the distances travelled by the drivers as a driving experience; as well as participation in traffic accidents were chosen. Four hundred questionnaire forms were given to the drivers. Among the given forms, 244 samples were filtered based on the answered questions. All of the incomplete and unreasonable answers in the form were invalidated. The filtered answered forms were composed of 143 male drivers and 101 female drivers. The ages of the drivers were separated into five groups: (18-25), (26-35), (36-45), (46-55), and (56-65) years old. The travel times were separated into five groups: (0-10), (10-20), (20-30), (30-40), and (40-50) hours a week. A statistical Software program (Minitab 16) was used to analyze the data regarding z-test for



comparisons and obtaining the correlations among the impulsiveness, aggressiveness, and demographic variables of the drivers.

#### 4. Results and Discussion

The correlations between impulsiveness and aggressiveness of the drivers are shown in Table 4. For the impulsiveness scale, the values of the two-factor and the one-factor structures were concerned. The second-order subscale analysis showed that attentional dimension moderately correlates with speeding ( $r_s = 0.449$ ), overt expression ( $r_s = 0.406$ ) and overall score of aggression ( $r_s = 0.472$ ); as well as with anger/aggression (0.340). The motor dimension scale correlates to the factors of ADBQ questionnaire: speeding ( $r_s = 0.476$ ), overall score of aggression ( $r_s = 0.424$ ), and anger/aggression ( $r_s = 0.327$ ). The non-planning dimension scale is less correlated with the ADBQ factors compared with the previous two second-order subscales; for example, the strongest two correlations were obtained with the speeding and overall score of aggression in which the  $r_s = 0.246$  and  $0.219$ , respectively.

With respect of the BIS-first-order subscales, attention dimension is positively correlated to the dimension of judgment of others ( $r_s = 0.323$ ), and with the total score on aggression scale ( $r_s = 0.34$ ). Cognitive instability is highly correlated with the total score on aggression scale ( $r_s = 0.423$ ), speeding ( $r_s = 0.404$ ), and overt expression ( $r_s = 0.363$ ). Also, first - order motor impulsiveness subscale is highly correlated with the speeding ( $r_s = 0.451$ ), total score on aggression scale ( $r_s = 0.416$ ), and anger/aggression ( $r_s = 0.337$ ). The perservance and self-control subscales are less correlated with the ADBQ questionnaire dimensions. The cognitive complexity subscale shows no significant correlation with any of the dimensions of aggression. The strongest correlation of the total score of impulsiveness are with speeding ( $r_s = 0.49$ ); then with the overall aggression score ( $r_s = 0.461$ ), and overt expression ( $r_s = 0.351$ ).

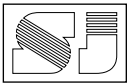
Regarding the correlation between the impulsiveness total score and overall aggressiveness score, it can be said that there is a moderate correlation is obtained between them, in which the  $r_s = 0.461$ .

Table 5 shows the correlation results among the impulsiveness, aggressiveness, and demographical variables that obtained from the questionnaire. The impulsiveness correlates with the age of the drivers negatively ( $r_s = -0.205$ ) and it correlates with the crashes positively ( $r_s = 0.157$ ); while it is not significantly correlated with gender, and hours driving per week. The aggressiveness of the drivers is correlated with crashes ( $r_s = 0.301$ ), with gender negatively ( $r_s = -0.197$ ), with age negatively ( $r_s = -0.177$ ), and with hours driving per week ( $r_s = 0.133$ ).

The comparisons of the impulsiveness first and second order factors between the male and female drivers are shown in Table 6. As indicated in the table, all of the p-values for all of the factors are greater than 0.05; therefore, it can be said that there are no significant differences between male and female drivers regarding the first and second order factors of the impulsiveness, as well as for the total score of the impulsivity of the drivers.

Despite of observing that there is no significant difference between male and female drivers for the impulsivity, Table 7 shows that there are significant differences between male and female drivers regarding Drivers' aggressiveness. Male drivers have higher values in anger/aggression, speeding, and overt expression aggressive dimension scales than female drivers. As well as, there is significant difference between male and female drivers for the total score of the aggressiveness. Because the P-value for the judgement of others scale is greater than 0.05; there are no significant difference between male and female drivers regarding this scale.

Table 8 shows the comparisons between the demographical variables of the drivers regarding the genders of the drivers. The table shows that there are significant differences between male and female drivers on number of crashes and travel



times of driving per week because the p-values are less than 0.05. Male drivers have faced higher accidents (1.86 crashes/ a driver) than female drivers (0.87 accidents/ a driver). Also, male drivers spend more times (13.81 hours/ week) compared to the female drivers (9.06 hours / week).

The aggressiveness and impulsiveness comparisons of the drivers regarding the age of the drivers are shown in Table 9. The aggressiveness average score of the drivers was decreased as the age of the drivers increased; for example, for the drivers age of 18 to 25 years old, the aggressive average score was 24.17, while for the 56 to 65 years old, the aggressiveness average score was 10.91. This result shows that traffic police should focus on younger drivers than elderly drivers, regarding enforcement and driving license. However, the impulsiveness average scores decreased when the age of the drivers increased, it can be said the differences between the scores are very small.

Table 10 shows the comparisons between the drivers' aggressiveness and impulsiveness regarding the hours that the drivers travel. The more the drivers travel, the more drivers' aggressiveness average scores obtained; for example, for the drivers who travel 0-10 hours a week, the aggressiveness average score was 16.74, while the average score was 24.31 for the drivers who travel 40-50 hours a week. There were no differences between the impulsiveness average scores of the drivers regarding the hours of the traveling per week.

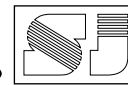
## 5. Conclusion

Some interesting results were obtained in this study as follows:

- Speeding as a measuring scale for the aggressive driving correlates more than other measuring scales with second-order impulsiveness subscales that are attentional, motor, and non-planning.
- The total score of the driving aggressiveness are more correlated with the attentional and

motor impulsiveness subscales than non-planning subscale.

- Among the first-order impulsiveness subscales, attention, cognitive, and motor subscales are significantly correlate with the measuring scales of aggressive driving, as well as with the total score of the aggressive driving.
- There is a moderate correlation between the impulsiveness total score and the overall aggressiveness score, in which  $r_s = 0.461$ .
- The impulsiveness of the drivers is negatively correlated with the drivers' ages and positively correlated with number of crashes.
- The aggressiveness of the drivers is significantly correlated with number of crashes and negatively with gender and age of the drivers.
- There are no significant differences between male and female drivers regarding the first and second order factors of the impulsiveness, as well as for the total score of the impulsivity of the drivers.
- Male drivers are more aggressive than female drivers in anger/aggression, speeding, and overt expression aggressive dimension scales, and in overall aggressive driving scale.
- Because male drivers are driving more aggressively than female drivers; therefore, they have higher crashes (1.86 crashes/ a driver) than female drivers. At the same time, male drivers spend more times in driving than female drivers.
- However, the aggressiveness average scores of the drivers were decreased as the age of the drivers increased, there were no big differences between the impulsiveness average scores.
- The more the drivers travel, the more drivers' aggressiveness average scores obtained; therefore, traffic police should focus more on the drivers that travel more than the drivers travel less.



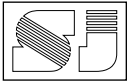
## 6. Recommendation

Based on the results of the study, the following points can be recommended that the traffic police in Sulaimaniyah City can concentrate on them:

- Speeding can be more observed by traffic police because it correlates more than other measuring scales with second-order impulsiveness subscales.
- The attentional and motor impulsiveness subscales can be more emphasized by traffic police during driving license process, because they are more correlated with the total score of the driving aggressiveness.
- Because younger drivers were more aggressive and impulsive than elderly drivers, the traffic police can observe them with more attention.
- Also, the traffic police should emphasize the aggressiveness scales of anger/aggression, speeding, and overt expression aggressive dimension scales during driving permission process especially for the male drivers.

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## بحث تحليلي للعدوانية والاندفاع وتصنيف السواق في مدينة السليمانية

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### المستخلص

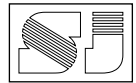
لغرض زيادة السلامة المرورية في طرقتنا تمت دراسة اشكال محددة من السلوك والسمات الشخصية للسواق السيارات الصغيرة. تم اجراء هذه الدراسة كمحاولة لفهم المساهمات المحتملة لاندفاعية وعدوانية السواق في حوادث السير في مدينة السليمانية. كما تم بحث العلاقة بين عدوانية واندفاعية السواق. عدد المشاركين في الدراسة الاستيعابية هو 244 سائق والذين ملأو استمارة مقياس اندفاع بارات (BTS-11) واستبيان سلوك السواق العدواني. عدد المشاركين الذكور كانو 143 سائقا بينما الاناث كانو 101 سائقا. نتائج التحليل الاحصائي اظهرت بان السواق الذكور يقودون مركباتهم بعدوانية اكثر من الاناث، ونتيجة لذلك فهم يواجهون عدد اكبر من حوادث السير مقارنة بالاناث. كما تبين عدم وجود فوارق ماحظة بينهما بخصوص اندفاعية السواق.

القيادة المسرعة كوحدة قياس لعدوانية السائق ترتبط بقوة مع الدرجة الثانية من المقياس الثانوي للاندفاعية. المقياس الثانوي للانتباه والاندفاعية الحركية يرتبط اكثر مع المجموع الكلي لنقاط عدوانية السائق مقارنة بالمقياس الثانوي غير المخطط.

هنالك علاقة معتدلة بين المجموع الكلي لنقاط اندفاع والنقاط الكاملة للعدوانية. اندفاعية السواق يرتبط بشكل عكسي مع عمر السائق وبشكل طردي مع عدد حوادث السير بينما عدوانية السواق ترتبط بشكل مؤثر مع عدد حوادث السير وبشكل عكسي مع جنس وعمر السائق.

**الكلمات المفتاحية:** العدوانية، الاندفاع، تصنيف السواق، سلامة المرور، سلوك السائق.



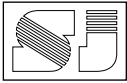
**Table 1: Barratt impulsiveness scale 11- factors structure and scoring** <sup>[27]</sup>

| 2 <sup>nd</sup> Order Factors | 1 <sup>st</sup> order Factors | No. of Items | Item contributing to each Subscale |
|-------------------------------|-------------------------------|--------------|------------------------------------|
| Attentional                   | Attention                     | 5            | 5, 9*, 11, 20*, 28                 |
|                               | Cognitive Instability         | 3            | 6, 24, 62                          |
| Motor                         | Motor                         | 7            | 2, 3, 4, 17, 19, 22, 25            |
|                               | Perseverance                  | 4            | 16, 21, 23, 30*                    |
| Non-planning                  | Self-Control                  | 6            | 1*, 7*, 8*, 12*, 13*, 14           |
|                               | Cognitive Complexity          | 5            | 10*, 15*, 18, 27, 29*              |

\* reverse scored items

**Table 2: ADBQ questionnaire form** <sup>[28]</sup>

| No. | Questions                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|
| 1.  | You become agitated or enraged when other drivers impede you, are not paying attention, or drive poorly around you on the road.   |
| 2.  | You travel above the speed limit, even if you have more than enough time to reach your destination.                               |
| 3.  | When other drivers do get on your nerves, how often do you think negatively of them without reacting verbally?                    |
| 4.  | You think that other drivers just are not thinking or paying enough attention when they anger you with their driving.             |
| 5.  | When other drivers annoy or anger you, you try to think positively or just accept there are frustrating situations while driving. |
| 6.  | In cases where you know you can get away with it, you have no problem breaking minor laws or rules.                               |
| 7.  | When another driver angers you while on the road, you follow very close (tailgate) or otherwise try to scare them.                |
| 8.  | You give the finger to drivers who annoy or anger you.                                                                            |
| 9.  | When another driver angers you while on the road, you shout verbal insults towards them, even if they cannot hear you.            |
| 10. | You stick your tongue out or make faces at drivers that annoy you or make you mad.                                                |
| 11. | You drive intoxicated even when you realize that you may be over the legal limit.                                                 |
| 12. | When another driver angers you at night, you shine your bright in their rearview mirror.                                          |
| 13. | You find being stuck in traffic or behind a slow driver especially annoying.                                                      |
| 14. | When another driver angers you while on the road, you attempt to get revenge on them.                                             |
| 15. | You find drivers that are impatient (ex. Weave in and out of traffic, disregard stop signs, etc.) especially annoying.            |
| 16. | While driving, you fail to notice signs or other cars, misjudge other's speed, etc.                                               |
| 17. | You "wake up" to realize that you have no clear recollection of the road along which you have just traveled.                      |
| 18. | You take chances and run through red lights.                                                                                      |
| 19. | If another driver is following too closely, you slow down or hit your breaks to get them to back off.                             |
| 20. | You shake your head at a driver who annoys you.                                                                                   |



**Table 3: Aggressive Driving Behavior Questionnaire factors <sup>[28]</sup>**

|          | Factor label                | Exemplar content                             |
|----------|-----------------------------|----------------------------------------------|
| Factor 1 | Anger/Aggression            | Tailgating, getting revenge on other drivers |
| Factor 2 | Speeding/ Minor infractions | Willingness to speed and break laws          |
| Factor 3 | Overt expression            | Sticking out tongue, giving the “finger”     |
| Factor 4 | Judgment of others          | Impatient drivers anger you                  |

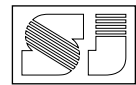
**Table 4: Correlations among Impulsiveness and Aggressiveness Driving Behavior Scales**

|                                      | 1. Attentional | 2. Motor | 3. Non-planning | 4. Attention | 5. Cognitive instability | 6. Motor | 7. Perservance | 8. Self-control | 9. Cognitive complexity | 10. Total score | 11. Anger/aggression | 12. Speeding | 13. Overt expression | 14. Judgment of others | 15. Total score |
|--------------------------------------|----------------|----------|-----------------|--------------|--------------------------|----------|----------------|-----------------|-------------------------|-----------------|----------------------|--------------|----------------------|------------------------|-----------------|
| <b>BIS-11 Second-order subscales</b> |                |          |                 |              |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 1                                    |                |          |                 |              |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 2                                    | .363           |          |                 |              |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 3                                    | .363           | .413     |                 |              |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| <b>BIS-11 First-order subscale</b>   |                |          |                 |              |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 4                                    | .845           | .383     | .383            |              |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 5                                    | .741           | .394     | .179            | .267         |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 6                                    | .476           | .904     | .407            | .370         | .394                     |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 7                                    | .233           | .622     | .196            | .195         | .176                     | .228     |                |                 |                         |                 |                      |              |                      |                        |                 |
| 8                                    | .363           | .440     | .886            | .383         | .174                     | .416     | .241           |                 |                         |                 |                      |              |                      |                        |                 |
| 9                                    | .190           | .167     | .868            | .193         | .099**                   | .191     | .029           | .271            |                         |                 |                      |              |                      |                        |                 |
| 10                                   | .745           | .818     | .783            | .652         | .525                     | .768     | .456           | .742            | .463                    |                 |                      |              |                      |                        |                 |
| <b>ADBQ</b>                          |                |          |                 |              |                          |          |                |                 |                         |                 |                      |              |                      |                        |                 |
| 11                                   | .299           | .327     | .171            | .189         | .303                     | .337     | .127*          | .195            | .049**                  | .334            |                      |              |                      |                        |                 |
| 12                                   | .449           | .476     | .246            | .323         | .404                     | .451     | .258           | .320            | .009**                  | .490            | .61                  |              |                      |                        |                 |
| 13                                   | .406           | .290     | .161*           | .293         | .363                     | .295     | .121**         | .225            | -.019**                 | .351            | .521                 | .667         |                      |                        |                 |
| 14                                   | .340           | .237     | .107**          | .269         | .274                     | .228     | .123**         | .149*           | -.012**                 | .278            | .395                 | .450         | .513                 |                        |                 |
| 15                                   | .472           | .424     | .219            | .340         | .423                     | .416     | .204           | .285            | .007**                  | .461            | .768                 | .874         | .844                 | .720                   |                 |

\* P-value < 0.05

\*\* P-value > 0.05

Others p-value < 0.01

**Table 5: Correlation among Impulsiveness, Aggressiveness, and Demographical Variables of Drivers**

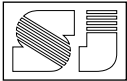
|                     | Impulsiveness | Aggressiveness | Gender  | Age     | Hours driving/<br>week | Crashes |
|---------------------|---------------|----------------|---------|---------|------------------------|---------|
| Impulsiveness       |               |                |         |         |                        |         |
| Aggressiveness      | 0.461         |                |         |         |                        |         |
| Gender              | -0.003**      | -0.197         |         |         |                        |         |
| Age                 | -0.205        | -0.177         | -0.131* |         |                        |         |
| Hours driving/ week | 0.016**       | 0.133*         | -0.260  | 0.073** |                        |         |
| Crashes             | 0.157*        | 0.301          | -0.29   | 0.166   | 0.127*                 |         |

**Table 6: Impulsiveness First and Second Order Factors Comparisons Regarding Gender of the Drivers**

|                       | Male  | S.D  | Female | S.D  | P-value |
|-----------------------|-------|------|--------|------|---------|
| Attention             | 36.7  | 14.5 | 39.9   | 14.2 | 0.095   |
| Cognitive instability | 37.4  | 19.8 | 36.3   | 18.3 | 0.664   |
| Attentional           | 37    | 13   | 38.5   | 12.9 | 0.356   |
| Motor                 | 32    | 15.4 | 31.6   | 15   | 0.848   |
| Perseverance          | 32.7  | 13.8 | 29.3   | 15.3 | 0.076   |
| Motor score           | 32.2  | 12.3 | 30.8   | 11.6 | 0.340   |
| Self-control          | 38.8  | 17.4 | 37     | 16.7 | 0.429   |
| Cognitive complexity  | 47.6  | 12.7 | 50.1   | 13.6 | 0.140   |
| Non-planning          | 42.8  | 12.4 | 43     | 12.4 | 0.901   |
| Impulsiveness         | 37.36 | 9.75 | 37.30  | 9.75 | 0.967   |

**Table 7: Aggressiveness Scale Comparisons Regarding Gender of the Drivers**

|                               | Male | S.D  | Female | S.D  | P-value |
|-------------------------------|------|------|--------|------|---------|
| Anger/Aggression              | 33   | 17.3 | 24.7   | 24.7 | 0.000   |
| Speeding                      | 21.2 | 18.8 | 15.4   | 13.4 | 0.005   |
| Overt expression              | 28.2 | 21.8 | 21.5   | 17.9 | 0.009   |
| Judgment of others            | 39.9 | 13.4 | 37.9   | 12.1 | 0.222   |
| Total score of aggressiveness | 30.6 | 14.6 | 25.1   | 10.8 | 0.001   |

**Table 8: Comparisons between the demographical variables of the drivers regarding drivers' genders**

|                              | Male  | S.D  | Female | S.D  | P-value |
|------------------------------|-------|------|--------|------|---------|
| Accidents (Accidents/Driver) | 1.86  | 1.92 | 0.87   | 1.02 | 0.000   |
| Travel Time (Hour/Week)      | 13.81 | 9.93 | 9.06   | 6.66 | 0.000   |

**Table 9: Average values of traffic aggressiveness and impulsiveness based on drivers' age**

| Score of aggressiveness and impulsiveness | Age range |       |       |       |       |
|-------------------------------------------|-----------|-------|-------|-------|-------|
|                                           | 18-25     | 26-35 | 36-45 | 46-55 | 56-65 |
| Number                                    | 50        | 77    | 57    | 39    | 21    |
| Average value of aggressiveness score     | 24.17     | 20.35 | 16.01 | 12.82 | 10.91 |
| Average value of impulsiveness score      | 38.38     | 40.07 | 36.41 | 33.59 | 34.29 |

**Table 10: Average value of traffic aggressiveness based on hours driving per week**

| Score of aggressiveness and impulsiveness | Hours per week range |       |       |       |       |
|-------------------------------------------|----------------------|-------|-------|-------|-------|
|                                           | 0-10                 | 10-20 | 20-30 | 30-40 | 40-50 |
| Number                                    | 114                  | 88    | 23    | 13    | 6     |
| Average value of aggressiveness score     | 16.74                | 18.23 | 19.93 | 23.08 | 24.31 |
| Average value of impulsiveness score      | 37.03                | 37.37 | 38.5  | 37.44 | 37.96 |